

# Zhexiang (Bruce) Li

M.S. Student in Computer Science, UCLA

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## EDUCATION

**University of California, Los Angeles** Sep 2026 - Jun 2028 (expected)

M.S. in Computer Science

**University of Southern California** May 2026

B.S. in Computer Science | GPA 3.76 / 4.00 | Viterbi Dean's List, Fall 2024-Spring 2026

## TECHNICAL SKILLS

**Languages** Python, C/C++, JavaScript/TypeScript, SQL

**ML and scientific computing** PyTorch, TensorFlow, Triton, CUDA, RDKit, Chemprop, recommender systems, multimodal inference

**Software and data** FastAPI, React, Next.js, PostgreSQL, Supabase, Docker, Linux

## RESEARCH EXPERIENCE

**Health Intelligence Lab (HAIL) | UCLA** Current

Research with Prof. Yuzhe Yang

Joined the lab; preparing to work on foundation models for type 1 diabetes (T1D).

**Machine Learning for Antiviral Drug Discovery | USC** Spring 2026

Research with Prof. Yan Liu's group

- Benchmarked RDKit, Chemprop, and hybrid models on seven TDC datasets with scaffold splits; ran feature-importance and SHAP analyses.

**Efficient Attention via Pre-Scoring | First author** 2024 - 2025

Advised by Prof. David P. Woodruff, Carnegie Mellon University | [arXiv:2505.11040](https://arxiv.org/abs/2505.11040)

- Proposed pre-scoring to select informative key intervals before attention; implemented Triton kernels and benchmarked against HyperAttention and LevAttention.

## INDUSTRY EXPERIENCE

**AI Engineer Intern | PMAOS / Next Big Seek** Jul - Aug 2026

- Designed a short-video recommender with bidirectional user-video interest propagation and separate session, bridge, and long-term exposure budgets. Added fatigue controls that cap repeated topics while reserving traffic for new interests.
- Built 11-language audio-visual affect inference with synchronization checks, presentation-attack detection, evidence fusion, and a fixed-point Q15 decision layer that can abstain or flag modality conflicts.
- Validated the recommendation logic with 629 tests and the multimodal pipeline with 139 Python tests, type and contract checks, and cross-compilation for Cortex-M3 and ARM926EJ-S.
- Evaluated a 9 mm piezoelectric millirobot concept and designed a PZT-steel cantilever experiment with acceptance criteria; work remained at the experimental planning stage.

**Full-Stack Engineer | Terlingua Threads** Jan - May 2026

- Built Shopify-integrated pages and Google Maps features; improved responsive layouts, accessibility, and SEO.

**Co-Founder and CTO | CrAlg Capital** 2025 - Mar 2026

- Built the data and workflow architecture for an AI-assisted housing debt-collection platform; set technical priorities and delivery standards.

## AI Engineer Intern | Piezolab Medical

Jun - Aug 2025

- Built an LLM-assisted pipeline to ingest, clean, and cluster 5,000+ medical-malpractice documents for quality-of-care review.
- Wrote experiment checklists and acceptance tests for vascular-care laboratory workflows.

## Application Engineer Intern | Synaptics

Jun - Aug 2023

- Validated product firmware and built an ESP32-to-Alibaba Cloud pipeline that published I2C sensor data as MQTT topics for internal dashboards.

## PUBLICATIONS AND MANUSCRIPTS

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### MedTokenBudget: Lesion-Preserving Token Routing for Dermoscopic Image Classification

Summer 2026

Lesion-aware token routing under fixed budgets, trained with mask supervision and evaluated for lesion retention on ISIC 2019. Submitted to AAAI 2027; [arXiv:2609.30613](#).

### Mass-Conservation Token Transformers for Auditable MS/MS Spectrum Retrieval

Summer 2026

Encoded mass-conserving fragment pairs as sparse tokens for spectrum reranking; tested ranking dependence with hard negatives and token-removal experiments. Submitted to AAAI 2027; preprint forthcoming.

### Efficient Attention via Pre-Scoring

2025

First author; preprint. [arXiv:2505.11040](#)

### Game Theory in Companies

2023

Modeled two-player release decisions and built a Python + Gambit pipeline to enumerate Nash equilibria. [DOI](#)

## SELECTED PROJECTS

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### Fridge Manager

2026

Built a Next.js and Supabase meal planner that uses inventory, leftovers, recipes, dietary preferences, and available time to produce meal and grocery plans.

### AI Business Card Hub

2026 - Present

Built an iOS organizer with vision parsing, OCR fallback, multi-photo review, structured people and company records, search, and bilingual display. [Code](#)

## ADDITIONAL EXPERIENCE

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### Investment Analyst Intern | GUDU Medical

Jan 2021 - Present

- Screen MedTech opportunities and support market research, due diligence, financial modeling, and valuation.